Untitled1

November 14, 2025

[1]:
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[2]:
```python
df = pd.read_excel(r'superstore_sales.xlsx')
```

[3]:
```python
df.head(5)
```

[3]:
```
          order_id order_date  ship_date       ship_mode    customer_name  \
0     AG-2011-2040 2011-01-01 2011-01-06  Standard Class  Toby Braunhardt
1    IN-2011-47883 2011-01-01 2011-01-08  Standard Class      Joseph Holt
2     HU-2011-1220 2011-01-01 2011-01-05    Second Class    Annie Thurman
3  IT-2011-3647632 2011-01-01 2011-01-05    Second Class     Eugene Moren
4    IN-2011-47883 2011-01-01 2011-01-08  Standard Class      Joseph Holt

       segment            state    country market   region  …  \
0     Consumer      Constantine    Algeria Africa   Africa  …
1     Consumer  New South Wales  Australia   APAC  Oceania  …
2     Consumer         Budapest    Hungary   EMEA     EMEA  …
3  Home Office        Stockholm     Sweden     EU    North  …
4     Consumer  New South Wales  Australia   APAC  Oceania  …

          category sub_category                 product_name    sales  \
0  Office Supplies      Storage          Tenex Lockers, Blue  408.300
1  Office Supplies     Supplies     Acme Trimmer, High Speed  120.366
2  Office Supplies      Storage       Tenex Box, Single Width   66.120
3  Office Supplies        Paper  Enermax Note Cards, Premium   44.865
4        Furniture  Furnishings   Eldon Light Bulb, Duo Pack  113.670

   quantity  discount   profit  shipping_cost order_priority  year
0         2       0.0  106.140          35.46         Medium  2011
1         3       0.1   36.036           9.72         Medium  2011
2         4       0.0   29.640           8.17           High  2011
3         3       0.5  -26.055           4.82           High  2011
4         5       0.1   37.770           4.70         Medium  2011

[5 rows x 21 columns]
```

[4]:
```
df.tail(5)
```

[4]:
```
            order_id  order_date   ship_date        ship_mode     customer_name  \
51285  CA-2014-115427  2014-12-31  2015-01-04  Standard Class        Erica Bern
51286    MO-2014-2560  2014-12-31  2015-01-05  Standard Class         Liz Preis
51287  MX-2014-110527  2014-12-31  2015-01-02    Second Class  Charlotte Melton
51288  MX-2014-114783  2014-12-31  2015-01-06  Standard Class     Tamara Dahlen
51289  CA-2014-156720  2014-12-31  2015-01-04  Standard Class     Jill Matthias

         segment              state        country  market   region  …  \
51285  Corporate         California  United States      US     West  …
51286   Consumer  Souss-Massa-Draâ         Morocco  Africa   Africa  …
51287   Consumer            Managua      Nicaragua   LATAM  Central  …
51288   Consumer          Chihuahua         Mexico   LATAM    North  …
51289   Consumer           Colorado  United States      US     West  …

              category sub_category  \
51285  Office Supplies      Binders
51286  Office Supplies      Binders
51287  Office Supplies       Labels
51288  Office Supplies       Labels
51289  Office Supplies    Fasteners

                                   product_name   sales  quantity  \
51285  Cardinal Slant-D Ring Binder, Heavy Gauge Vinyl  13.904         2
51286          Wilson Jones Hole Reinforcements, Clear   3.990         1
51287           Hon Color Coded Labels, 5000 Label Set  26.400         3
51288           Hon Legal Exhibit Labels, Alphabetical   7.120         1
51289                              Bagged Rubber Bands   3.024         3

       discount   profit  shipping_cost  order_priority  year
51285       0.2   4.5188          0.890          Medium  2014
51286       0.0   0.4200          0.490          Medium  2014
51287       0.0  12.3600          0.350          Medium  2014
51288       0.0   0.5600          0.199          Medium  2014
51289       0.2  -0.6048          0.170          Medium  2014

[5 rows x 21 columns]
```

[5]:
```
# Shape of the dataset
df.shape
```

[5]: (51290, 21)

[6]:
```
# Columns present in the dataset
df.columns
```

[6]:

```
Index(['order_id', 'order_date', 'ship_date', 'ship_mode', 'customer_name',
       'segment', 'state', 'country', 'market', 'region', 'product_id',
       'category', 'sub_category', 'product_name', 'sales', 'quantity',
       'discount', 'profit', 'shipping_cost', 'order_priority', 'year'],
      dtype='object')
```

[7]:

```
#A concise summary of the dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   order_id        51290 non-null  object
 1   order_date      51290 non-null  datetime64[ns]
 2   ship_date       51290 non-null  datetime64[ns]
 3   ship_mode       51290 non-null  object
 4   customer_name   51290 non-null  object
 5   segment         51290 non-null  object
 6   state           51290 non-null  object
 7   country         51290 non-null  object
 8   market          51290 non-null  object
 9   region          51290 non-null  object
 10  product_id      51290 non-null  object
 11  category        51290 non-null  object
 12  sub_category    51290 non-null  object
 13  product_name    51290 non-null  object
 14  sales           51290 non-null  float64
 15  quantity        51290 non-null  int64
 16  discount        51290 non-null  float64
 17  profit          51290 non-null  float64
 18  shipping_cost   51290 non-null  float64
 19  order_priority  51290 non-null  object
 20  year            51290 non-null  int64
dtypes: datetime64[ns](2), float64(4), int64(2), object(13)
memory usage: 8.2+ MB
```

[8]:

```
# Checking Missing values present
df.isnull().sum()
```

[8]:

```
order_id          0
order_date        0
ship_date         0
ship_mode         0
customer_name     0
segment           0
```

```
state             0
country           0
market            0
region            0
product_id        0
category          0
sub_category      0
product_name      0
sales             0
quantity          0
discount          0
profit            0
shipping_cost     0
order_priority    0
year              0
dtype: int64
```

[9]:

```python
# Getting descriptive statistics summary
df.describe()
```

[9]:

```
                          order_date                      ship_date  \
count                          51290                          51290
mean   2013-05-11 21:26:49.155780864  2013-05-15 20:42:42.745174528
min              2011-01-01 00:00:00            2011-01-03 00:00:00
25%              2012-06-19 00:00:00            2012-06-23 00:00:00
50%              2013-07-08 00:00:00            2013-07-12 00:00:00
75%              2014-05-22 00:00:00            2014-05-26 00:00:00
max              2014-12-31 00:00:00            2015-01-07 00:00:00
std                              NaN                            NaN

              sales      quantity      discount        profit  shipping_cost  \
count  51290.000000  51290.000000  51290.000000  51290.000000   51290.000000
mean     246.490581      3.476545      0.142908     28.641740      26.375818
min        0.444000      1.000000      0.000000  -6599.978000       0.002000
25%       30.758625      2.000000      0.000000      0.000000       2.610000
50%       85.053000      3.000000      0.000000      9.240000       7.790000
75%      251.053200      5.000000      0.200000     36.810000      24.450000
max    22638.480000     14.000000      0.850000   8399.976000     933.570000
std      487.565361      2.278766      0.212280    174.424113      57.296810

               year
count  51290.000000
mean    2012.777208
min     2011.000000
25%     2012.000000
50%     2013.000000
75%     2014.000000
```

```
max      2014.000000
std         1.098931
```

[10]:
```python
df['order_date'].min()
```

[10]: `Timestamp('2011-01-01 00:00:00')`

[11]:
```python
df['order_date'].max()
```

[11]: `Timestamp('2014-12-31 00:00:00')`

[24]:
```python
df['order_date'] = pd.to_datetime(df['order_date'])
```

[25]:
```python
df['month_year'] = df['order_date'].dt.to_period('M').astype(str)
```

[26]:
```python
df_trend = df.groupby('month_year')['sales'].sum().reset_index()
```

[27]:
```python
plt.figure(figsize=(15,6))
plt.plot(df_trend['month_year'], df_trend['sales'], color='#b80045')
plt.xticks(rotation='vertical', size=8)
plt.show()
```

[34]:
```python
prod_sales=pd.DataFrame(df.groupby('product_name')['sales'].sum())
```

[38]:
```python
prod_sales = prod_sales.sort_values('sales',ascending=False)
```

[39]:
```python
prod_sales[:10]
```

[39]:
```
                                                            sales
product_name
```

```
Apple Smart Phone, Full Size                        86935.7786
Cisco Smart Phone, Full Size                        76441.5306
Motorola Smart Phone, Full Size                     73156.3030
Nokia Smart Phone, Full Size                        71904.5555
Canon imageCLASS 2200 Advanced Copier               61599.8240
Hon Executive Leather Armchair, Adjustable          58193.4841
Office Star Executive Leather Armchair, Adjustable  50661.6840
Harbour Creations Executive Leather Armchair, A…  50121.5160
Samsung Smart Phone, Cordless                       48653.4600
Nokia Smart Phone, with Caller ID                   47877.7857
```

[44]:
```
most_sell_prod=pd.DataFrame(df.groupby('product_name').
↪sum(numeric_only=True)['quantity'])
```

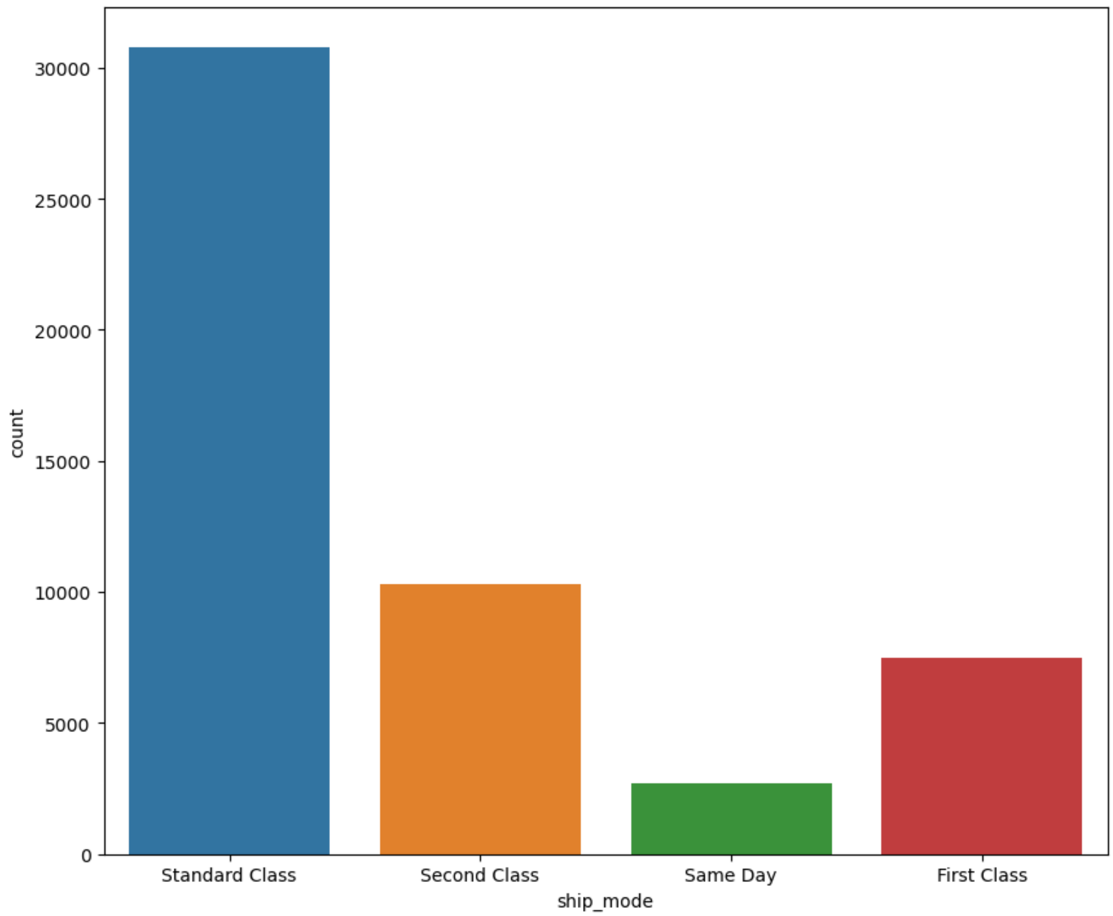
[46]:
```
most_sell_prod=most_sell_prod.sort_values('quantity',ascending=False)
```

[47]:
```
most_sell_prod[:10]
```

[47]:

```
                                       quantity
product_name
Staples                                     876
Cardinal Index Tab, Clear                   337
Eldon File Cart, Single Width               321
Rogers File Cart, Single Width              262
Sanford Pencil Sharpener, Water Color       259
Stockwell Paper Clips, Assorted Sizes       253
Avery Index Tab, Clear                      252
Ibico Index Tab, Clear                      251
Smead File Cart, Single Width               250
Stanley Pencil Sharpener, Water Color       242
```

[50]:
```
plt.figure(figsize=(10,8.5))
sns.countplot(x='ship_mode', data=df)
plt.show()
```



[55]:
```
cat_subcat_profit=pd.DataFrame(df.groupby(['category','sub_category']).
    ↪sum(numeric_only=True)['profit'])
```

[57]:
```
cat_subcat_profit.sort_values(['category','profit'],ascending=False)
```

[57]:

category	sub_category	profit
Technology	Copiers	258567.54818
	Phones	216717.00580
	Accessories	129626.30620
	Machines	58867.87300
Office Supplies	Appliances	141680.58940
	Storage	108461.48980
	Binders	72449.84600
	Paper	59207.68270
	Art	57953.91090
	Envelopes	29601.11630
	Supplies	22583.26310

```
                Labels          15010.51200
                Fasteners       11525.42410
Furniture       Bookcases      161924.41950
                Chairs         141973.79750
                Furnishings     46967.42550
                Tables         -64083.38870
```

```
[ ]:
```